From Board to Graph: A Grid-Based World Model for Hybrid Human–AI Role-Play that Makes Narrative Power Visible

[Anonymised for review]

Abstract

Whose story becomes the story of a community depends on who is where, who knows what and who controls the places everyone needs. Games can make these forces tangible. We present the design of a world model for role-playing in which the world is a board, represented as a matrix, and in which human players and generative-AI characters can share the same table. The model connects three graphs: the spatial graph of the board, the temporal graph of encounters between characters and the causal graph of the events they produce. A character can only build on events it witnessed or heard about, so the board decides which stories can meet. The design responds to an empirical finding: when language-model agents share an unrestricted world, their storylines merge into a single story within days, pulled by "attractors" written into the premises. A proof of concept with synthetic agents shows that space and information alone only delay this convergence, that attractor places decide where it happens, and that two simple mechanics, world-event cards and a focus rule, keep several stories alive. We describe a physical kit in which causal links are drawn with yarn between event cards, the hybrid human–AI protocol, and a research plan to test whether playing on such a board helps groups see how space, information and power shape the stories a community ends up telling.

1 Introduction

Every community tells itself stories about what happened: who was responsible for a death, why the grain ran out, who is to blame for the debt. Which of those stories survive is rarely a matter of truth alone. It depends on who was present, on who heard what from whom, and on who controls the places where people meet: the warehouse, the customs house, the mill. Stories told by people at the periphery of these networks are absorbed into the dominant account or never reach it. Games for social change aim to make such processes experienceable, so that players can reflect on them (Bogost, 2007; Flanagan, 2009; Flanagan & Nissenbaum, 2014).

Role-playing games are a natural medium for this. Their stories emerge from what the participants do in a shared world (Fine, 1983; Montola, 2008), and they are already used to develop perspective-taking and collaborative skills (Bowman, 2010; Daniau, 2016). Generative AI now makes it possible to populate these worlds with artificial characters (Park et al., 2023) and to mix human and AI players at the same table. This raises a design question with a social side: when humans and AI characters co-create a community's story, what decides which stories survive?

A companion study, currently under review, gives a first answer for the case in which every character is an AI agent. Language-model agents acting freely in a small town produced four to five storylines on the second day of play, and within a few days these storylines merged into one. Storylines almost

never ended; they were absorbed. The path of the convergence was not random: the first fusions joined characters whose premises (their objectives, secrets and beliefs) already referred to each other, and they happened in the places those premises named. We call such premises and places *attractors*, borrowing the concept from dynamical systems (Strogatz, 2015). One feature of that simulator matters here: a single model call decided what every character did and saw everything that had happened recently. The characters were, in effect, omniscient.

This paper takes the next step on the design side. We propose a world model in which the world is a *board*, with the properties that board games make explicit: positions, distances, paths, regions and limited vision (Salen & Zimmerman, 2004; Woods, 2012). On this board, a character only knows what it saw or was told, and can only build on what it knows. The board is represented as a matrix, and its structure propagates, through two other graphs, to the structure of the story. The same board can be played by AI agents in a simulator and by people with a physical kit, which allows human, AI and hybrid tables to be compared with the same instruments.

The contributions are: (1) a formal model linking the board, the encounters between characters and the causal graph of the story; (2) a proof of concept with synthetic agents that identifies which board mechanics change convergence and which do not; (3) a physical kit and a hybrid human–AI protocol; and (4) a research plan, framed as a game for critical thinking about how communities come to share one version of events.

2 Background

Space as narrative architecture. Game spaces tell stories by arranging places, paths and obstacles that channel movement and encounters (Jenkins, 2004). A game's fiction is constrained by its rules, and its rules are often spatial (Juul, 2005; Salen & Zimmerman, 2004). Modern board games make these structures particularly legible: the map is visible to everyone, and distances and adjacencies are part of the rules (Woods, 2012). Architecture offers a precedent for measuring this: space syntax describes buildings and cities as graphs and shows that the position of a space in that graph predicts how much it is used and how people meet in it (Hillier & Hanson, 1984).

Grids as models of social dynamics. Grids have a long history as minimal models of social processes. Schelling's model showed that mild individual preferences on a grid produce strong segregation (Schelling, 1971); Sugarscape grew whole artificial societies on a grid of resources (Epstein & Axtell, 1996); cellular automata showed how local rules on a lattice produce complex global patterns (Wolfram, 1984). These models are also games in a loose sense, and they have been used to teach exactly the point we care about: that collective outcomes can emerge from local structure without anyone intending them.

Stories as graphs. Narrative comprehension can be modelled as a causal network of events (Trabasso & van den Broek, 1985), and the kernels of a plot are the events with causal antecedents and consequences (Barthes & Duisit, 1975; Chatman, 1978). Network science supplies the tools to read such graphs: connected components, bridges and articulation points (Tarjan, 1972), communities (Blondel et al., 2008) and the conditions under which a growing random graph becomes connected (Newman, 2018). Emergent narrative research has long noted that autonomous agents produce events more easily than stories with shape (Aylett, 1999; Louchart & Aylett, 2004; Riedl & Bulitko, 2013).

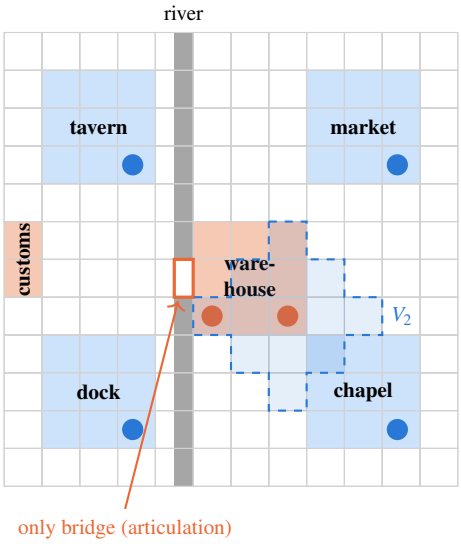


Figure 1: An example board: places are regions of cells; the warehouse and the customs house are attractor places, named in several premises; a river with a single bridge creates an articulation cell. Discs are characters; the shaded cells are what one character sees with perception radius 2 (cells at most two steps away).

3 The model

3.1 The board as a matrix

The board is a grid of $n \times n$ cells. Each cell is a passage, a barrier or part of a *place*; a place p is a connected region of cells with a name and a description (the warehouse, the customs house, the chapel). Figure 1 shows an example. Two non-barrier cells are adjacent when they share a side (the 4-neighbourhood; hexagonal and 8-neighbour boards fit the same framework). The board is therefore a graph $S = (C, A)$ whose vertices are the cells, and it is fully described by its adjacency matrix $A \in \{0,1\}^{|C|\times|C|}$, with $A_{xy} = 1$ when cells x and y are adjacent.

Matrix powers answer the board's basic questions. A cell y is reachable from x in at most k steps when

$$\left[(I+A)^k\right]_{xy} > 0, \tag{1}$$

so a character that moves m cells per turn can reach, in one turn, the cells in the support of row x of $(I+A)^m$. Perception uses the same operator: a character at x with perception radius ρ sees what happens in the cells $V_\rho(x) = \{y : [(I+A)^\rho]_{xy} > 0\}$. Distances, bottlenecks and chokepoints are properties of S: a bridge of S is a passage whose removal disconnects the board, and an articulation cell is a cell whose removal does so (Tarjan, 1972). The places that many premises mention are the *attractor places*; their *spatial basin* is the set of cells from which characters tend to move toward them.

3.2 Characters, encounters and knowledge

Each character a has a position $x_a(t)$ at turn t, a home place, and premises: an objective, a secret and beliefs, as in the companion study. Characters move toward the places their premises make relevant. Two characters *meet* at turn t when one is within the other's perception radius. Encounters form a temporal graph H over the characters: let $M^{(t)} \in \{0,1\}^{|\mathscr{A}|\times|\mathscr{A}|}$ be the encounter matrix at turn t. Information can

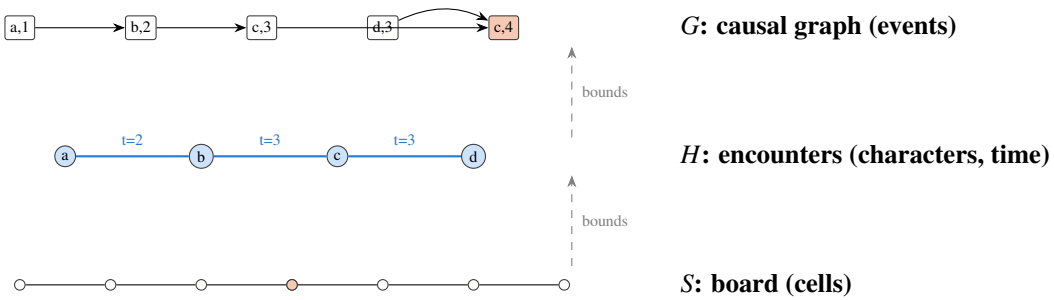


Figure 2: The three layers. Where characters can go (S) bounds whom they meet and when (H), which bounds what each event can cite (G). The highlighted event, at an attractor place, merges two storylines.

travel from a to b between turns t_0 and t_1 only along a time-respecting path, which exists when

$$\Big[\prod_{t=t_0}^{t_1-1}\big(I+M^{(t)}\big)\Big]_{ab} > 0, \tag{2}$$

where the product is taken in Boolean arithmetic. Each character holds a *knowledge set* $K_a(t)$: the events it caused, the events it saw (those in $V_\rho(x_a)$) and the events it heard about from characters it met, each passed on with a rumour probability π.

3.3 Events and the causal graph

Each turn, each character acts. The action becomes an event v with an author, a cell, a turn and a list of declared causes, exactly as in the existing simulator. The constraint that defines the model is

$$u \in \text{causes}(v) \;\Rightarrow\; u \in K_{\text{author}(v)}\big(t(v)\big), \tag{3}$$

that is, a character can only build on what it knows. For AI characters the constraint is enforced by construction: each character is prompted separately and the prompt lists only its knowledge set, instead of the single omniscient call used so far. For human players it is enforced by the physical kit (Section 5).

The causal graph G is built from the declared causes, and storylines are its connected components, recomputed every day, with births, fusions and closures tracked as in the companion study. The three graphs are nested: S bounds H (characters can only meet where the board lets them go), and H bounds G (an event can only cite another if information about it could have travelled to its author). This yields design predictions that can be read directly off the board. Attractor places should concentrate the fusions; bridges and articulation cells of S should be where separate storylines first meet; and storylines rooted far apart on the board should merge later. Figure 2 summarises the three layers.

4 Proof of concept with synthetic agents

Before involving language models or people, we asked which board mechanics can change convergence at all. We wrote a minimal simulation in which six synthetic characters move on a 12×12 board with six places, two of them attractor places, and act once per day for 20 days. Each event cites a number of causes drawn from the distribution observed with language-model agents in the companion study (mostly one or two causes; almost never none), chosen at random among the recent events the author *knows*. The characters have no content and no intentions: what is tested is the structure alone. We compared the

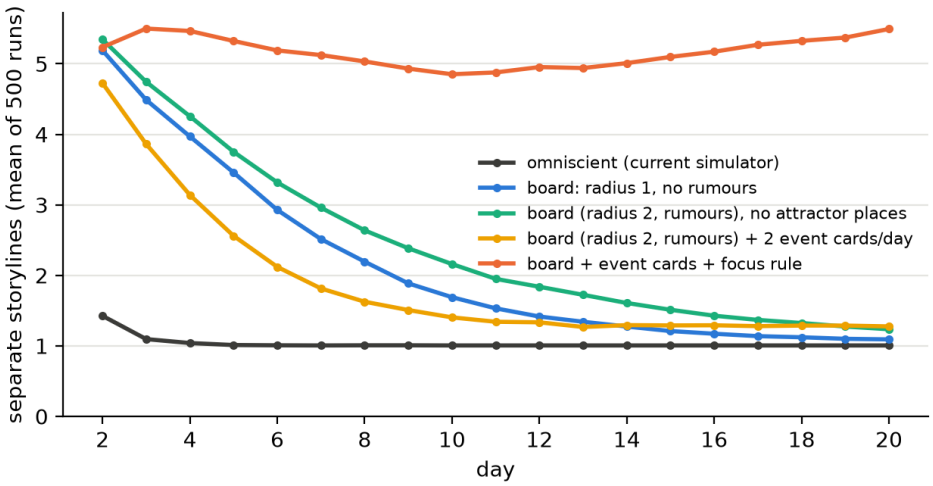


Figure 3: Separate storylines per day in the synthetic proof of concept (mean of 500 runs per condition, 20 days).

Table 1: Proof of concept, 20 days, 500 runs per condition.

Condition	Storylines at the end	Runs ending with one	Born after day 2	Fusions near attractors
Omniscient (current simulator)	1.01	99%	0.1	91%
Board, radius 2, rumours	1.05	95%	0.0	85%
Board, radius 1, no rumours	1.09	91%	0.0	85%
Board, radius 2, rumours, no attractor places	1.24	79%	0.0	66%
Board + 2 event cards per day	1.28	76%	1.3	86%
Board + event cards + focus rule	5.49	2%	6.0	84%

omniscient regime of the existing simulator with board regimes that vary the perception radius, rumours and attractor places, and with two design levers: *event cards*, two world events per day that appear in random cells with no cause, and a *focus rule*, under which, in 80% of actions, all causes must come from the same storyline. Each condition was run 500 times; the script is in the repository.

Three results stand out (Figure 3, Table 1). First, *information changes the beginning, not the end.* In the omniscient regime, the actions of the second day already cite each other across characters and only 1.5 storylines are born on average; on the board, each character mostly knows its own past, and about five storylines are born, the same order observed with language-model agents. But on every board without further mechanics, almost all runs still end with a single storyline. Reducing the perception radius or removing rumours only slows the process.

Second, *attractor places decide where, and partly whether.* With attractor places, more than 80% of fusions happen next to them; without them, the share falls to 66% and one run in five keeps more than one storyline.

Third, *new stories need new seeds and a reason not to knot.* Since every character always knows its own past and almost always cites something, no storyline is born after the second day unless the world supplies events without causes. Event cards provide such seeds, but alone they are soon absorbed. Combined with the focus rule, which keeps an action inside one storyline, they sustain five to six storylines throughout the session, with new ones being born as others merge.

The proof of concept has obvious limits: synthetic characters choose at random and have no premises in the textual sense. Its value is to separate what the board can do from what content does, and to point

at the mechanics worth testing with people and language models.

5 The physical kit and the hybrid protocol

The same board is printed for a table of four to eight participants. The kit has: the board, with places and a river; one token and one character sheet per character, with objective, secret and beliefs; blank *event cards*, on which a player writes what the character does, the turn and the identifier of each cause; coloured yarn, which connects each event card to the cards it cites; a deck of world-event cards; and screens that hide each player's known cards.

A turn has four steps. (1) Characters move up to m cells. (2) The facilitator reveals world-event cards in the drawn cells. (3) Each player writes an event card and places it on the cell where it happens; the card's causes must be cards the character knows, and the facilitator ties the yarn. (4) Knowledge is updated: a player adds to their hand the cards placed within the perception radius of their token, and players whose tokens meet may show each other cards from their hands (rumours). Optional rules switch the focus rule on or off, or change ρ and the number of world-event cards.

In hybrid tables, some characters are played by the simulator. The facilitator, or a companion app, enters the AI characters' positions and known cards; the simulator, prompted per character with only that knowledge, returns the action and its causes, which are written on cards like any other. Because every event card carries the same fields as the simulator's event log, a photographed or transcribed table yields the same data as a simulated session, and all analyses (storylines, fusions, attractors) run unchanged on human, AI and hybrid tables.

At the end of a session the yarn *is* the causal graph. The debrief uses it directly: which storylines merged, at which card, in which place; which characters' storylines were absorbed into others and never had their own ending; who knew what, and when.

6 Research plan

We plan two studies, to be pre-registered with the same procedures as the companion study (Nosek et al., 2018).

Study 1 (simulation with language-model agents). The existing simulator will be extended with the board and with per-character prompts restricted to knowledge sets. With the same worlds and models as the companion study, we will cross the omniscient and board regimes, the perception radius, rumours, event cards and the focus rule. Hypotheses: (H1) on the board, more storylines are born and fusions happen later than in the omniscient regime; (H2) fusions concentrate in attractor places and at articulation cells of the board; (H3) without event cards, no storylines are born after the first days in any regime; (H4) event cards with the focus rule sustain several storylines and allow closures by stability.

Study 2 (tables with people). Workshops with groups of students and community participants will play the same scenario in three formats: human only, AI only (projected) and hybrid. Each group plays twice, with and without the board constraints. Measures include the same graph metrics, plus a short post-game reflection in which participants describe how the town's story was formed and who shaped it, coded for references to information, place and power. (H5) Human tables cite fewer causes per event than AI agents and keep more storylines separate; (H6) after the debrief with the yarn graph, participants refer more often to structural factors (who knew what, who controlled which place) when explaining the outcome of the story. Studies with people will follow the ethics procedures of the authors' institution, with informed consent and anonymised data.

7 Relevance for social change

The board is a small model of a social question: how a community comes to share one version of events. In the companion study, the story converged around the places and secrets the world's authors had entangled; in this model, the same happens around places that control movement and around characters who meet many others. Players who see their character's storyline absorbed into someone else's, because they were on the wrong side of the river or never heard the rumour, experience, rather than are told, how access to information and to central places translates into narrative power. The debrief makes that experience explicit with an object everyone built together, the yarn graph.

This connects to three themes of games for social change. *Critical thinking*: the game asks why a story became the story, not only what happened. *Civic awareness*: the scenarios (a harbour town with debt, bribery and smuggling; a village rationing grain before winter) are about commons, corruption and scarcity, and the board shows how place and information shape collective decisions about them. *Inclusion*: the mechanics show how peripheral voices are absorbed, and the levers that change it (event cards that seed new stories, a focus rule that protects a thread, more bridges across the river) are design choices players can discuss and change. Because the same board is played by AI characters, it also opens a conversation about how generative AI, which in our data links everything it sees, may amplify the convergence toward a single account.

8 Limitations and next steps

The model is a design proposal with a proof of concept, not an evaluated game. The proof of concept uses synthetic characters and says nothing about the content language models or people would produce on the board. The formal model treats knowledge as sets of events, which is a simplification of what characters believe and misremember. The physical kit has not yet been playtested; its turn structure may be too heavy for some groups, and a digital companion may be needed to manage knowledge sets. The next steps are the board module of the simulator, a first playtest of the kit, and the pre-registration of the two studies.

9 Conclusion

Representing the world of a role-playing game as a board, and the board as a matrix, makes it possible to reason about the story with the tools of graph theory: the board bounds the encounters, the encounters bound the causal graph, and the causal graph is the story. A proof of concept shows that the board alone does not prevent a shared story from collapsing into one, that attractor places decide where it collapses, and that two simple mechanics keep several stories alive. The same board, played by people and AI characters with yarn and cards, turns this structure into something a group can see and discuss: how space, information and power decide which story a community ends up telling.

References

Aylett, R. (1999). Narrative in virtual environments: Towards emergent narrative. *Narrative Intelligence: Papers from the AAAI Fall Symposium*, (FS-99-01), 83–86.

Barthes, R., & Duisit, L. (1975). An introduction to the structural analysis of narrative. *New Literary History*, *6*(2), 237. https://doi.org/10.2307/468419

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, *2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/p10008

Bogost, I. (2007). *Persuasive games: The expressive power of videogames*. MIT Press.

Bowman, S. L. (2010). *The functions of role-playing games: How participants create community, solve problems and explore identity*. McFarland.

Chatman, S. (1978). *Story and discourse: Narrative structure in fiction and film*. Cornell University Press.

Daniau, S. (2016). The transformative potential of role-playing games: From play skills to human skills. *Simulation & Gaming*, *47*(4), 423–444. https://doi.org/10.1177/1046878116650765

Epstein, J. M., & Axtell, R. L. (1996). *Growing artificial societies: Social science from the bottom up*. The MIT Press. https://doi.org/10.7551/mitpress/3374.001.0001

Fine, G. A. (1983). *Shared fantasy: Role-playing games as social worlds*. University of Chicago Press.

Flanagan, M. (2009). *Critical play: Radical game design*. The MIT Press. https://doi.org/10.7551/mitpress/7678.001.0001

Flanagan, M., & Nissenbaum, H. (2014). *Values at play in digital games*. The MIT Press. https://doi.org/10.7551/mitpress/9016.001.0001

Hillier, B., & Hanson, J. (1984). *The social logic of space*. Cambridge University Press. https://doi.org/10.1017/cbo9780511597237

Jenkins, H. (2004). Game design as narrative architecture. In N. Wardrip-Fruin & P. Harrigan (Eds.), *First person: New media as story, performance, and game* (pp. 118–130). MIT Press.

Juul, J. (2005). *Half-real: Video games between real rules and fictional worlds*. MIT Press.

Louchart, S., & Aylett, R. (2004). Narrative theory and emergent interactive narrative. *International Journal of Continuing Engineering Education and Lifelong Learning*, *14*(6), 506. https://doi.org/10.1504/ijceell.2004.006017

Montola, M. (2008). The invisible rules of role-playing: The social framework of role-playing process. *International Journal of Role-Playing*, (1), 22–36. https://doi.org/10.33063/ijrp.vi1.184

Newman, M. (2018). *Networks* (2nd ed.). Oxford University Press. https://doi.org/10.1093/oso/9780198805090.001.0001

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, *115*(11), 2600–2606. https://doi.org/10.1073/pnas.1708274114

Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 1–22. https://doi.org/10.1145/3586183.3606763

Riedl, M. O., & Bulitko, V. (2013). Interactive narrative: An intelligent systems approach. *AI Magazine*, *34*(1), 67–77. https://doi.org/10.1609/aimag.v34i1.2449

Salen, K., & Zimmerman, E. (2004). *Rules of play: Game design fundamentals*. MIT Press.

Schelling, T. C. (1971). Dynamic models of segregation†. *The Journal of Mathematical Sociology*, *1*(2), 143–186. https://doi.org/10.1080/0022250x.1971.9989794

Strogatz, S. H. (2015). *Nonlinear dynamics and chaos: With applications to physics, biology, chemistry, and engineering* (2nd ed.). Westview Press.

Tarjan, R. (1972). Depth-first search and linear graph algorithms. *SIAM Journal on Computing*, *1*(2), 146–160. https://doi.org/10.1137/0201010

Trabasso, T., & van den Broek, P. (1985). Causal thinking and the representation of narrative events. *Journal of Memory and Language*, *24*(5), 612–630. https://doi.org/10.1016/0749-596x(85)90049-x

Wolfram, S. (1984). Cellular automata as models of complexity. *Nature*, *311*(5985), 419–424. https://doi.org/10.1038/311419a0

Woods, S. (2012). *Eurogames: The design, culture and play of modern european board games*. McFarland.